

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems
COURSE CODE: CSA0509

COURSE OUTCOME COVERED

CO2: *Apply the relational model, relational algebra, SQL query structures, and views to solve database querying and data manipulation problems. (BL3, BL6)*

Activity Tool : Debugging & Optimisation Task (Low)
Assessment Tool Weightage: 10%

QUESTIONS		
Q.N o	Question	Bloom's Level
1	Identify and correct errors in a given SQL CREATE TABLE statement with incorrect syntax and missing constraints.	BL2 – Understand
2	Debug an SQL query that incorrectly uses aggregate functions without GROUP BY and fix the issue. Bloom's Level:	BL2 – Understand
3	Correct a faulty SELECT query where column names are misspelled or improperly referenced.	BL1 – Remember
4	Fix errors in an SQL query using JOIN, where the join condition is missing or incorrect.	BL3 – Apply
5	Identify and correct a subquery error where multiple values are returned instead of a single value.	BL2 – Understand
6	Debug a VIEW creation statement that contains invalid column references or syntax errors.	BL1 – Remember
7	Examine an SQL query with unnecessary conditions and optimize it by simplifying the WHERE clause.	BL3 – Apply
8	Identify violations of integrity constraints in a given table and correct them.	BL2 – Understand

9	Fix a relation that violates 1NF or 2NF by removing repeating groups or partial dependencies.	BL3 – Apply
10	Identify redundant attributes in a relation and apply basic normalization (up to 3NF) to improve design.	BL3 – Apply

QUESTION:1 Identify and correct errors in a given SQL CREATE TABLE statement with incorrect syntax and missing constraints.

ANSWERS:

```
mysql> CREATE DATABASE DebugDB;
Query OK, 1 row affected (0.01 sec)
```

```
mysql> USE DebugDB;
Database changed
```

ERROR:

```
mysql> CREATE TABLE Student_Info student_id INT, student_name VARCHAR, marks, PRIMARY KEY student_id;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'student_id INT, student_name VARCHAR, marks, PRIMARY KEY student_id' at line 1
```

CORRECTED:

```
mysql> CREATE TABLE Student_Info (student_id INT PRIMARY KEY, student_name VARCHAR(100), marks DECIMAL(5,2));
Query OK, 0 rows affected (0.04 sec)
```

QUESTION: 2 Debug an SQL query that incorrectly uses aggregate functions without GROUP BY and fix the issue. Bloom's Level:

ANSWERS:

```
mysql> CREATE TABLE Staff (staff_id INT PRIMARY KEY, staff_name VARCHAR(100), dept_name VARCHAR(50), salary DECIMAL(10,2));
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> INSERT INTO Staff VALUES (1,'John','HR',40000),(2,'Alice','HR',45000),(3,'Bob','IT',60000),(4,'David','IT',65000),(5,'Eva','Sales',50000);
Query OK, 5 rows affected (0.03 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

ERROR:

```
mysql> SELECT dept_name, AVG(salary) FROM Staff;
ERROR 1140 (42000): In aggregated query without GROUP BY, expression #1 of SELECT list contains nonaggregated column 'debugdb.Staff.dept_name'; this is incompatible with sql_mode=only_full_group_by
```

CORRECTED:

```
mysql> SELECT dept_name, AVG(salary) FROM Staff GROUP BY dept_name;
+-----+-----+
| dept_name | AVG(salary) |
+-----+-----+
| HR        | 42500.000000 |
| IT        | 62500.000000 |
| Sales     | 50000.000000 |
+-----+-----+
3 rows in set (0.01 sec)
```

QUESTION: 3 Correct a faulty SELECT query where column names are misspelled or improperly referenced.

ANSWERS:

```
mysql> CREATE TABLE Employee (emp_id INT PRIMARY KEY, emp_name VARCHAR(100), department VARCHAR(50), salary DECIMAL(10,2));
Query OK, 0 rows affected (0.02 sec)

mysql> INSERT INTO Employee VALUES (1,'Rahul','HR',35000),(2,'Priya','IT',50000),(3,'Arun','Sales',42000);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

ERROR:

```
mysql> SELECT emp_nam, departmnt, salry FROM Employee;
ERROR 1054 (42S22): Unknown column 'emp_nam' in 'field list'
```

CORRECTED:

```
mysql> SELECT emp_name, department, salary FROM Employee;
+-----+-----+-----+
| emp_name | department | salary |
+-----+-----+-----+
| Rahul    | HR         | 35000.00 |
| Priya    | IT         | 50000.00 |
| Arun     | Sales      | 42000.00 |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

QUESTION: 4 Fix errors in an SQL query using JOIN, where the join condition is missing or incorrect.

ANSWERS:

```
mysql> CREATE TABLE Department (dept_id INT PRIMARY KEY, dept_name VARCHAR(50));
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO Department VALUES (1,'HR'),(2,'IT'),(3,'Sales');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT*FROM Department;
+-----+-----+
| dept_id | dept_name |
+-----+-----+
|      1 | HR       |
|      2 | IT       |
|      3 | Sales    |
+-----+-----+
3 rows in set (0.00 sec)

mysql> ALTER TABLE Staff ADD dept_id INT;
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> UPDATE Staff SET dept_id=1 WHERE staff_id=1;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE Staff SET dept_id=1 WHERE staff_id=2;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE Staff SET dept_id=2 WHERE staff_id=3;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

ERROR:

```
mysql> SELECT s.staff_name,d.dept_name FROM Staff s JOIN Department d ON s.department_id=d.dept_id;
ERROR 1054 (42S22): Unknown column 's.department_id' in 'on clause'
```

CORRECTED:

```
mysql> SELECT s.staff_name,d.dept_name FROM Staff s JOIN Department d ON s.dept_id=d.dept_id;
+-----+-----+
| staff_name | dept_name |
+-----+-----+
| John      | HR       |
| Alice     | HR       |
| Bob       | IT       |
| David     | IT       |
| Eva       | Sales    |
+-----+-----+
5 rows in set (0.00 sec)
```

QUESTION: 5 Identify and correct a subquery error where multiple values are returned instead of a single value.

ANSWERS:

```
mysql> CREATE TABLE Departments (dept_id INT PRIMARY KEY, dept_name VARCHAR(50));
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> INSERT INTO Departments VALUES (1,'HR'),(2,'IT'),(3,'Sales');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> CREATE TABLE Staffs (staff_id INT PRIMARY KEY, staff_name VARCHAR(100), dept_name VARCHAR(50), salary DECIMAL(10,2), dept_id INT);
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> INSERT INTO Staff VALUES (1,'John','HR',40000,1),(2,'Alice','HR',45000,1),(3,'Bob','IT',60000,2),(4,'David','IT',65000,2),(5,'Eva','Sales',50000,3);
ERROR 1062 (23000): Duplicate entry '1' for key 'staff.PRIMARY'
mysql> INSERT INTO Staffs VALUES (1,'John','HR',40000,1),(2,'Alice','HR',45000,1),(3,'Bob','IT',60000,2),(4,'David','IT',65000,2),(5,'Eva','Sales',50000,3);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Staffs;
```

staff_id	staff_name	dept_name	salary	dept_id
1	John	HR	40000.00	1
2	Alice	HR	45000.00	1
3	Bob	IT	60000.00	2
4	David	IT	65000.00	2
5	Eva	Sales	50000.00	3

```
5 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Departments;
```

dept_id	dept_name
1	HR
2	IT
3	Sales

```
3 rows in set (0.00 sec)
```

ERROR:

```
mysql> SELECT staff_name FROM Staff WHERE dept_id=(SELECT dept_id
FROM Department);
ERROR 1242 (21000): Subquery returns more than 1 row
```

CORRECTED:

```
mysql> SELECT staff_name FROM Staff WHERE dept_id IN (SELECT dept
_id FROM Department);
+-----+
| staff_name |
+-----+
| John      |
| Alice     |
| Bob       |
| David     |
| Eva       |
+-----+
5 rows in set (0.00 sec)
```

QUESTION: 6 Debug a VIEW creation statement that contains invalid column references or syntax errors.

ANSWERS:**ERROR:**

```
mysql> CREATE VIEW Staff_View AS SELECT s.staff_name,d.department
FROM Staff s JOIN Department d ON s.dept_id=d.dept_id;
ERROR 1054 (42S22): Unknown column 'd.department' in 'field list'
```

CORRECTED:

```
mysql> CREATE VIEW Staff_View AS SELECT s.staff_name,d.dept_name
FROM Staff s JOIN Department d ON s.dept_id=d.dept_id;
Query OK, 0 rows affected (0.02 sec)

mysql> SELECT * FROM Staff_View;
+-----+-----+
| staff_name | dept_name |
+-----+-----+
| John      | HR        |
| Alice     | HR        |
| Bob       | IT        |
| David     | IT        |
| Eva       | Sales     |
+-----+-----+
5 rows in set (0.00 sec)
```

QUESTION: 7 Examine an SQL query with unnecessary conditions and optimize it by simplifying the WHERE clause.

ANSWERS:

ORIGINAL (UNOPTIMIZED):

```
mysql> SELECT * FROM Staff WHERE salary>50000 AND salary>=30000 AND dept_name='IT' AND dept_name='IT';
```

staff_id	staff_name	dept_name	salary	dept_id
3	Bob	IT	60000.00	2
4	David	IT	65000.00	2

```
2 rows in set (0.00 sec)
```

CORRECTED (OPTIMIZED):

```
mysql> SELECT * FROM Staff WHERE salary>50000 AND dept_name='IT';
```

staff_id	staff_name	dept_name	salary	dept_id
3	Bob	IT	60000.00	2
4	David	IT	65000.00	2

```
2 rows in set (0.00 sec)
```

QUESTION: 8 Identify violations of integrity constraints in a given table and correct them.

ANSWERS:

```
mysql> INSERT INTO Orders VALUES (101,NULL,'Completed');
ERROR 1146 (42S02): Table 'debugdb.orders' doesn't exist
mysql> CREATE TABLE Orders (order_id INT PRIMARY KEY, customer_id INT NOT NULL, status VARCHAR(20) CHECK(status IN('Pending','Shipped','Delivered')));
Query OK, 0 rows affected (0.04 sec)
```

ERROR:

```
mysql> INSERT INTO Orders VALUES (101,NULL,'Completed');
ERROR 1048 (23000): Column 'customer_id' cannot be null
```

CORRECTED:

```
mysql> INSERT INTO Orders VALUES (101,1,'Pending');
Query OK, 1 row affected (0.01 sec)

mysql> SELECT*FROM Orders;
```

order_id	customer_id	status
101	1	Pending

```
1 row in set (0.00 sec)
```

QUESTION: 9 Fix a relation that violates 1NF or 2NF by removing repeating groups or partial dependencies

ANSWERS:

ERROR:

```
mysql> CREATE TABLE Employee_Project (emp_id INT, emp_name VARCHAR(100), project1 VARCHAR(50), project2 VARCHAR(50), project3 VARCHAR(50));
ERROR 1050 (42S01): Table 'employee_project' already exists
mysql> CREATE TABLE Employee_Project (emp_id INT, emp_name VARCHAR(100), project1 VARCHAR(50), project1 VARCHAR(50));
ERROR 1050 (42S01): Table 'employee_project' already exists
```

CORRECTED:

```
mysql> CREATE TABLE Employee_Projects (emp_id INT, emp_name VARCHAR(100), project1 VARCHAR(50), project2 VARCHAR(50), project3 VARCHAR(50));
Query OK, 0 rows affected (0.04 sec)
```

QUESTION: 10 Identify redundant attributes in a relation and apply basic normalization (up to 3NF) to improve design.

ANSWERS:

```
mysql> CREATE TABLE Order_Details (order_id INT, customer_name VARCHAR(100), customer_city VARCHAR(50), product_name VARCHAR(100), product_price DECIMAL(10,2), quantity INT);
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> CREATE TABLE Customer (customer_id INT PRIMARY KEY, customer_name VARCHAR(100), customer_city VARCHAR(50));
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> CREATE TABLE Customer (customer_id INT PRIMARY KEY, customer_name VARCHAR(100), customer_city VARCHAR(50));
ERROR 1050 (42S01): Table 'customer' already exists
mysql> CREATE TABLE Product (product_id INT PRIMARY KEY, product_name VARCHAR(100), product_price DECIMAL(10,2));
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> CREATE TABLE Orders (order_id INT PRIMARY KEY, customer_id INT, FOREIGN KEY(customer_id) REFERENCES Customer(customer_id));
```

```
ERROR 1050 (42S01): Table 'orders' already exists
mysql> CREATE TABLE Order_Item (order_id INT, product_id INT, quantity INT, PRIMARY KEY(order_id, product_id), FOREIGN KEY(order_id) REFERENCES Orders(order_id), FOREIGN KEY(product_id) REFERENCES Product(product_id));
Query OK, 0 rows affected (0.05 sec)
```



```
mysql> INSERT INTO Customer VALUES (1,'Rahul','Chennai'),(2,'Priya','Coimbatore');
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Customer;
+-----+-----+-----+
| customer_id | customer_name | customer_city |
+-----+-----+-----+
|          1 | Rahul         | Chennai       |
|          2 | Priya         | Coimbatore    |
+-----+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> INSERT INTO Product VALUES (101,'Laptop',55000.00),(102,'Mouse',800.00),(103,'Keyboard',1500.00);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Product;
+-----+-----+-----+
| product_id | product_name | product_price |
+-----+-----+-----+
|          101 | Laptop       | 55000.00      |
|          102 | Mouse        | 800.00         |
|          103 | Keyboard     | 1500.00        |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

Code of Conduct:

I _ M o n i k a . R (1 9 2 3 2 4 2 8 1) _ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Monika.R

RUBRICS FOR CALCULATION:

Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation